

```
In [1]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import warnings
warnings.filterwarnings("ignore")
```

C:\Users\Prashant\anaconda3\lib\site-packages\pandas\core\arrays\masked.py:60: UserWarning: Pandas requires version '1.3.6' or newer of 'bottleneck' (version '1.3.5' currently installed).  
from pandas.core import i

```
In [2]: df = pd.read_csv('placement.csv')
```

```
In [3]: df.head()
```

```
Out[3]:
```

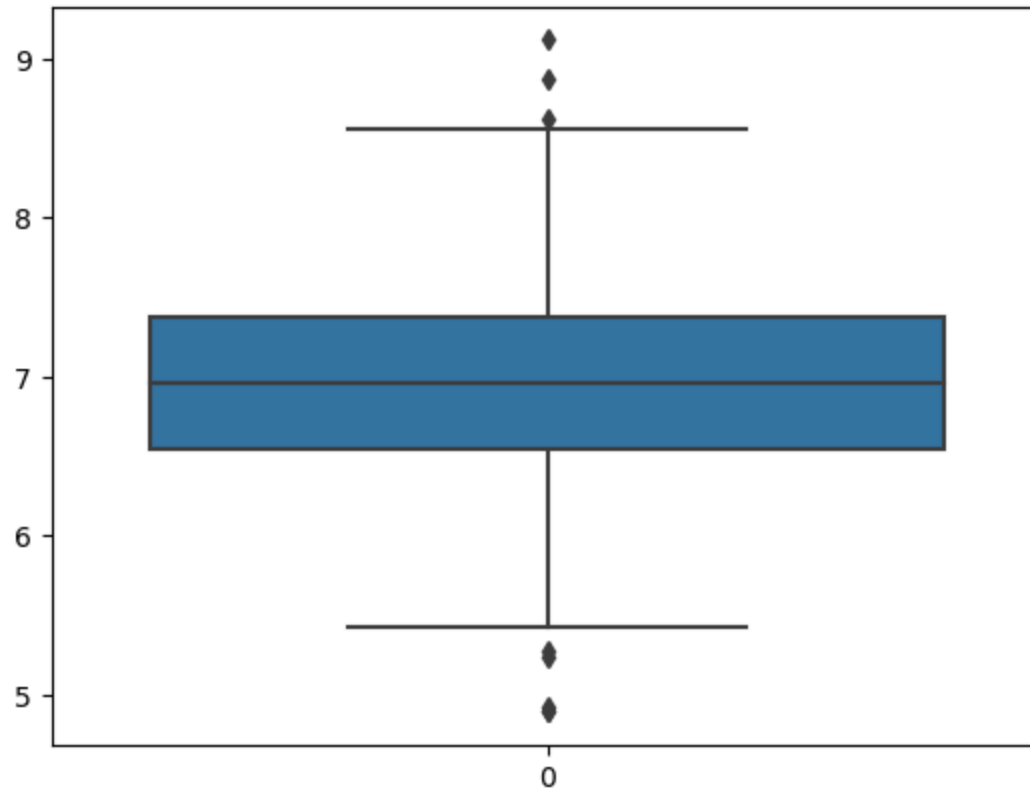
|   | cgpa | placement_exam_marks | placed |
|---|------|----------------------|--------|
| 0 | 7.19 | 26.0                 | 1      |
| 1 | 7.46 | 38.0                 | 1      |
| 2 | 7.54 | 40.0                 | 1      |
| 3 | 6.42 | 8.0                  | 1      |
| 4 | 7.23 | 17.0                 | 0      |

```
In [4]: df.shape
```

```
Out[4]: (1000, 3)
```

```
In [5]: sns.boxplot(df["cgpa"])
```

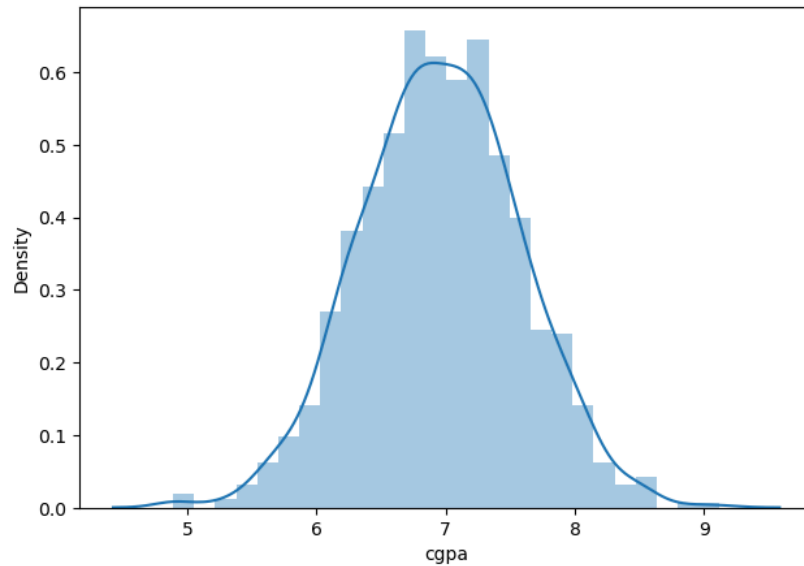
```
Out[5]: <Axes: >
```



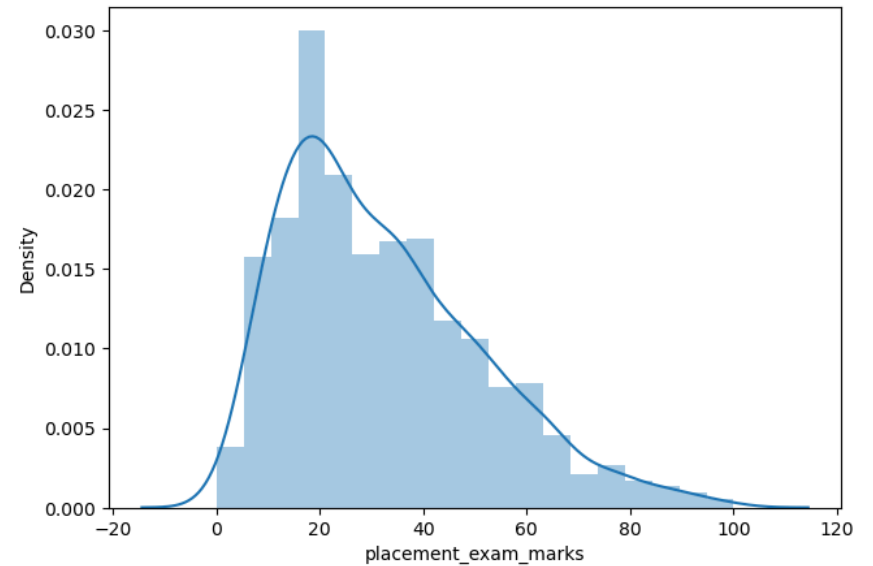
```
In [6]: plt.figure(figsize=(16,5))
plt.subplot(1,2,1)
sns.distplot(df['cgpa'])

plt.subplot(1,2,2)
sns.distplot(df['placement_exam_marks'])

plt.show()
```



Outlier with IQR



```
In [7]: # Median
#median_cgpa = np.median(df['cgpa'])
#print("Median :",median_cgpa)

# Lower quartile
q1_cgpa = df['cgpa'].quantile(0.25)

# upper quartile
q3_cgpa = df['cgpa'].quantile(0.75)

# printing Results
print("Q1:", q1_cgpa)
print("Q3:", q3_cgpa)
print("IQR:", q3_cgpa - q1_cgpa)
```

```
Q1: 6.55
Q3: 7.37
IQR: 0.8200000000000003
```

```
In [8]: ## Lets find no. of outliers

# for that we have to find the upper and ower outlier limit
outlier_lower_limit = q1_cgpa - 1.5*(q3_cgpa - q1_cgpa)
outlier_upper_limit = q3_cgpa + 1.5*(q3_cgpa - q1_cgpa)
print("Outlier lower Limit :", outlier_lower_limit)
print("Outlier upper Limit :", outlier_upper_limit)
```

Outlier lower Limit : 5.319999999999999  
Outlier upper Limit : 8.600000000000001

```
In [9]: df[df.cgpa<outlier_lower_limit]
```

```
Out[9]:
```

|     | cgpa | placement_exam_marks | placed |
|-----|------|----------------------|--------|
| 278 | 5.23 | 17.0                 | 1      |
| 485 | 4.92 | 44.0                 | 1      |
| 815 | 5.27 | 43.0                 | 0      |
| 997 | 4.89 | 34.0                 | 0      |
| 999 | 4.90 | 10.0                 | 1      |

```
In [10]: df[df.cgpa>outlier_upper_limit]
```

```
Out[10]:
```

|     | cgpa | placement_exam_marks | placed |
|-----|------|----------------------|--------|
| 995 | 8.87 | 44.0                 | 1      |
| 996 | 9.12 | 65.0                 | 1      |
| 998 | 8.62 | 46.0                 | 1      |

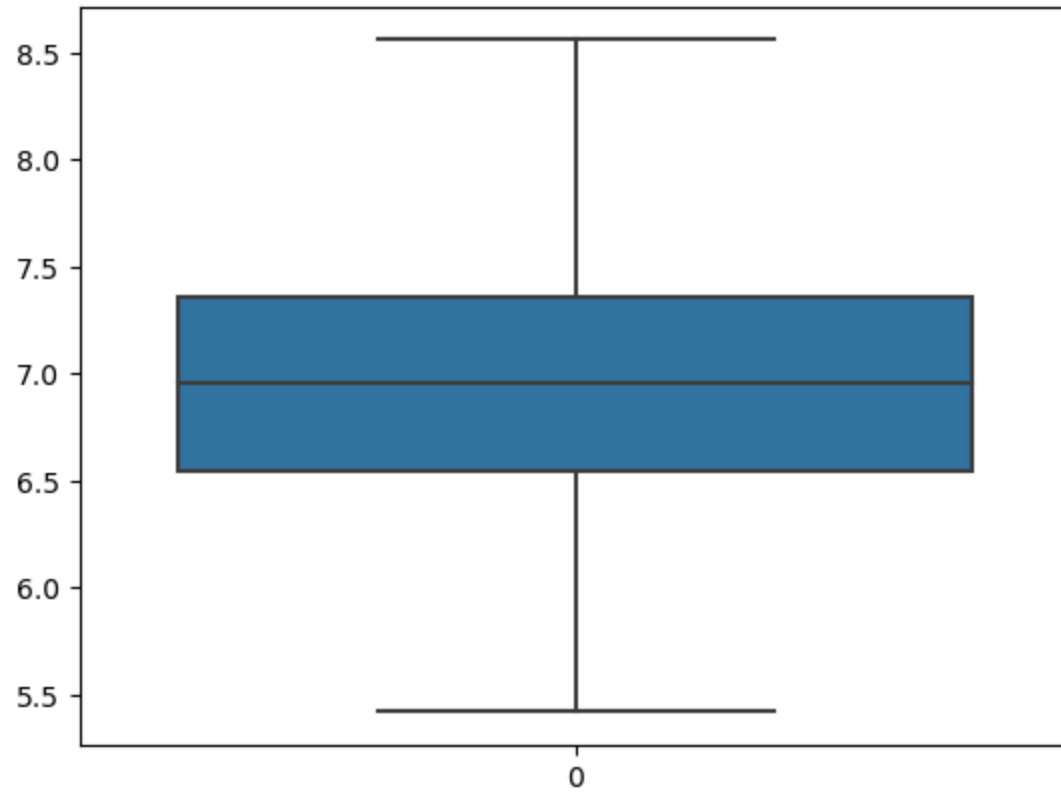
```
In [11]: data_df=df[(df.cgpa>outlier_lower_limit) & (df.cgpa<outlier_upper_limit)]
```

```
In [12]: data_df.shape
```

```
Out[12]: (992, 3)
```

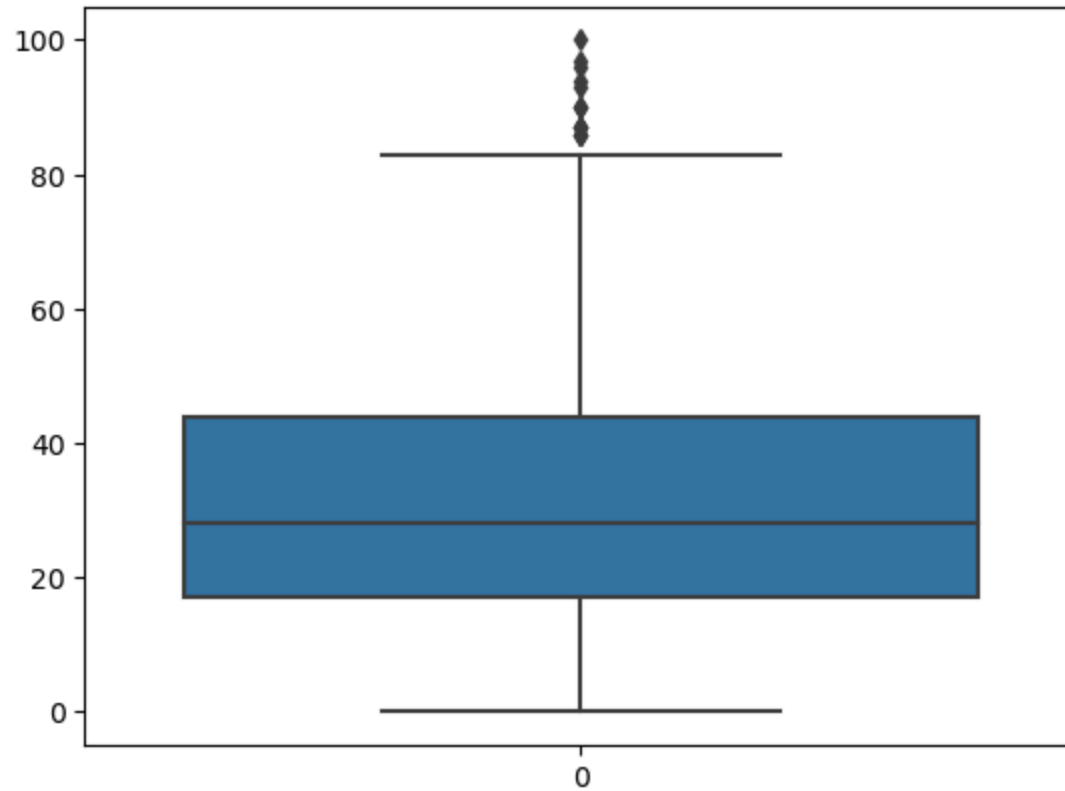
```
In [8]: sns.boxplot(data_df["cgpa"])
```

```
Out[8]: <Axes: >
```



In [ ]:

In [9]: `sns.boxplot(df['placement_exam_marks'])`Out[9]: `<Axes: >`



```
In [35]: df['placement_exam_marks'].describe()
```

```
Out[35]: count    1000.000000
mean       32.225000
std        19.130822
min         0.000000
25%        17.000000
50%        28.000000
75%        44.000000
max        100.000000
Name: placement_exam_marks, dtype: float64
```

```
In [22]: # Finding the IQR
q1 = df['placement_exam_marks'].quantile(0.25)
q3 = df['placement_exam_marks'].quantile(0.75)
```

```
In [11]: q3
```

Out[11]: 44.0

```
In [24]: iqr = q3 - q1
```

```
In [13]: iqr
```

Out[13]: 27.0

```
In [25]: upper_limit = q3 + 1.5 * iqr  
lower_limit = q1 - 1.5 * iqr
```

```
In [26]: print("Upper limit",upper_limit)  
print("Lower limit",lower_limit)
```

Upper limit 84.5  
Lower limit -23.5

## Finding Outliers

```
In [44]: df[df['placement_exam_marks'] > upper_limit]
```

Out[44]:

|     | cgpa | placement_exam_marks | placed |
|-----|------|----------------------|--------|
| 9   | 7.75 | 94.0                 | 1      |
| 40  | 6.60 | 86.0                 | 1      |
| 61  | 7.51 | 86.0                 | 0      |
| 134 | 6.33 | 93.0                 | 0      |
| 162 | 7.80 | 90.0                 | 0      |
| 283 | 7.09 | 87.0                 | 0      |
| 290 | 8.38 | 87.0                 | 0      |
| 311 | 6.97 | 87.0                 | 1      |
| 324 | 6.64 | 90.0                 | 0      |
| 630 | 6.56 | 96.0                 | 1      |
| 685 | 6.05 | 87.0                 | 1      |
| 730 | 6.14 | 90.0                 | 1      |
| 771 | 7.31 | 86.0                 | 1      |
| 846 | 6.99 | 97.0                 | 0      |
| 917 | 5.95 | 100.0                | 0      |

```
In [ ]: df[df['placement_exam_marks'] < lower_limit]
```

## Trimming

```
In [28]: new_df = df[df['placement_exam_marks'] < upper_limit]
```

```
In [29]: new_df.shape
```

```
Out[29]: (985, 3)
```

```
In [48]: # Comparing
plt.figure(figsize=(16,8))
```

```
plt.subplot(2,2,1)
sns.distplot(df['placement_exam_marks'])

plt.subplot(2,2,2)
sns.boxplot(df['placement_exam_marks'])

plt.subplot(2,2,3)
sns.distplot(new_df['placement_exam_marks'])

plt.subplot(2,2,4)
sns.boxplot(new_df['placement_exam_marks'])

plt.show()
```

